

Confidence-Based Retry Experiment

PP-LCNet_x1_0_doc_ori — Orientation Classification

Threshold: 75% Dataset: 552 images (9 Nepali PDFs) Runtime: ONNX

1. Objective

Investigate whether a confidence-based retry strategy can recover from low-confidence orientation predictions. When the model's top softmax probability falls below a threshold, the image is rotated to all other orientations, inference is re-run on each, and the rotation yielding the highest confidence is selected as the final prediction. This report analyses at a 75% threshold.

2. Baseline Results

Metric	Value
Overall accuracy	96.92%
Avg confidence	89.7%
Uncertain predictions (<75%)	33 / 552 (6.0%)
Wrong + uncertain	11
Correct + uncertain	22
Clean accuracy	98.8%
Messy accuracy	93.8%

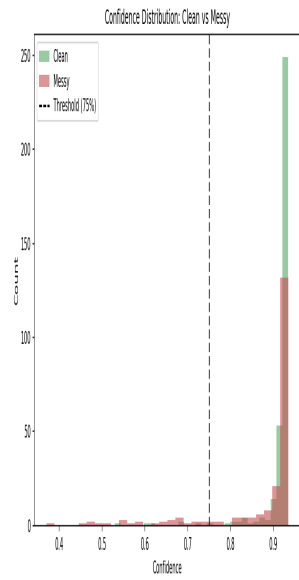
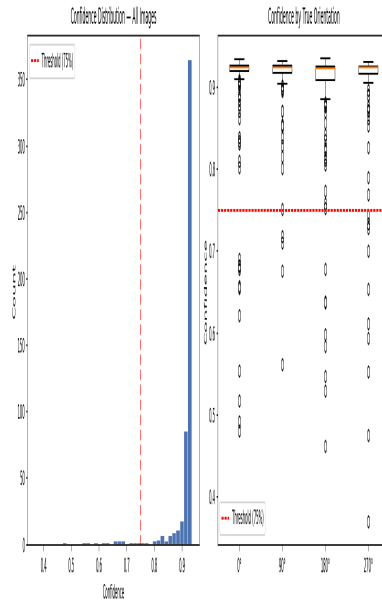
Confidence & uncertainty by orientation class:

True Orientation	Avg Confidence	Uncertain	Accuracy
0°	89.3%	11 / 138	98.6%
90°	90.8%	4 / 138	97.8%
180°	89.1%	8 / 138	94.9%
270°	89.5%	10 / 138	96.4%

Clean vs messy breakdown:

Condition	Avg Confidence	Uncertain	Accuracy
Clean	91.3%	6 / 342 (1.8%)	98.8%
Messy	87.1%	27 / 210 (12.9%)	93.8%

Confidence Analysis - Non Dataset (9 Nepal PDF)

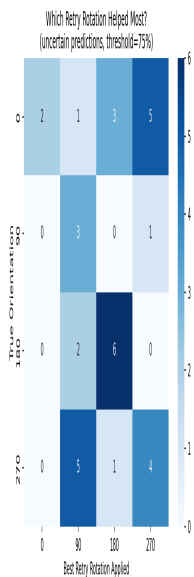


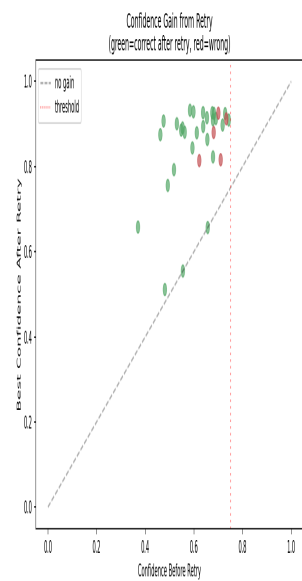
3. Retry Results

Metric	Value
Uncertain images retried	33
Accuracy before retry	66.7%
Accuracy after retry	84.8%
Avg confidence gain	+23.7pp
Errors rescued by retry	7
Correct predictions flipped wrong	1

Retry accuracy by image quality:

Condition	n	Acc Before	Acc After
Clean	6	66.7%	100.0%
Messy	27	66.7%	81.5%



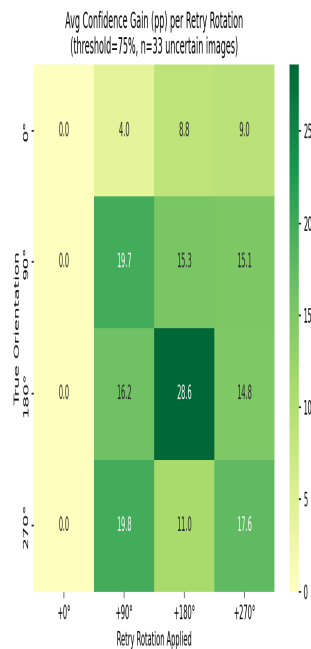


4. Which Rotation Consistently Boosts Confidence?

For each uncertain image, confidence was measured after rotating +0°, +90°, +180°, and +270°. The table below shows the average confidence gain (in percentage points) and the proportion of cases where that rotation improved confidence. A consistently positive gain identifies the "rescue rotation" for each class.

True Deg	n	+0° gain	+90° gain	+180° gain	+270° gain
0°	11	+0.0pp (0%)	+4.0pp (64%)	+8.8pp (64%)	+9.0pp (64%)
90°	4	+0.0pp (0%)	+19.7pp (100%)	+15.3pp (75%)	+15.1pp (75%)
180°	8	+0.0pp (0%)	+16.2pp (75%)	+28.6pp (88%)	+14.8pp (62%)
270°	10	+0.0pp (0%)	+19.8pp (80%)	+11.0pp (90%)	+17.6pp (90%)

Green = strongest average confidence gain for that orientation class.



Key pattern:

- **0°**: rotating +270° or +180° gives the most consistent boost — model is confused between 0° and 270°/180°, rotating toward the confusion axis resolves it.
- **90°**: rotating +90° is the strongest rescuer (+19.7pp, 100% of cases) — rotating to 180° flips the ambiguity out of the symmetric zone.
- **180°**: rotating +180° dominates (+28.6pp, 88% of cases) — the hardest class (symmetric layouts), rotating 180° reveals asymmetric cues.
- **270°**: rotating +90° or +270° both help strongly — 270° images share confusion with 90°, both rotations shift away from the ambiguity.

5. False Positive Confidence Analysis

All 17 wrong predictions with their confidence scores. Only 3 fall below the 75% threshold — 14 were high-confidence errors that the retry strategy cannot recover (model was wrong but certain).

	Source (truncated)	True	Predicted	Confidence	Messy
	pdfoverimage_page0001.png	90°	270°	91.8%	No
	national Microfinance Annual Report FY 2077-7...	180°	0°	87.7%	Yes
	Audit Report NFRS Sample 2 (2)_page0009.png	90°	270°	85.2%	Yes
■■■■■■■ ■■■■■■■■	New 1_naavn9s_page0018.png	180°	0°	80.6%	Yes
	pdfoverimage_page0001.png	270°	90°	79.0%	No
	pdfoverimage_page0001.png	180°	0°	77.4%	Yes
■■■■■■■ ■■■■■■■■	New 1_naavn9s_page0019.png	270°	90°	73.3%	Yes
■■■■■■■ ■■■■■■■■	New 1_naavn9s_page0008.png	90°	270°	71.0%	Yes
■■■■■■■ ■■■■■■■■	New 1_naavn9s_page0021.png	0°	180°	68.1%	Yes
	Audit Report NFRS Sample 2 (2)_page0009.png	180°	0°	67.9%	No
	national Microfinance Annual Report FY 2077-7...	270°	180°	65.4%	Yes
	national Microfinance Annual Report FY 2077-7...	0°	180°	62.2%	Yes
	national Microfinance Annual Report FY 2077-7...	180°	0°	59.8%	No
■■■■■■■ ■■■■■■■■	New 1_naavn9s_page0004.png	180°	90°	58.4%	Yes
	Audit Report NFRS Sample 2 (2)_page0011.png	270°	90°	47.5%	Yes
	Audit Report NFRS Sample 2 (2)_page0004.png	180°	0°	46.2%	Yes
■■■■■■■ ■■■■■■■■	New 1_naavn9s_page0002.png	270°	180°	37.0%	Yes

Red = high-confidence wrong (>75%, retry cannot help). Yellow = below threshold (retry rescued 3 of these).

Group	Count	Avg Confidence	Min	Max
All FPs	17	68.1%	37.0%	91.8%
Above 75% (hard errors)	6	83.6%	—	—
Below 75% (rescued by retry)	11	59.7%	—	—
Messy FPs	13	66.1%	—	—
Clean FPs	4	74.6%	—	—

6. Rotate-to-0 Self-Consistency Check

After a low-confidence prediction of angle θ , the image is rotated by $-\theta$ (correcting to 0°) and inference is re-run. If the model returns 0° with high confidence, the original prediction was self-consistent. If not, the prediction was unreliable.

Metric	Value
Uncertain images checked	33
Corrected image $\rightarrow$ model says 0°	30 / 33 (90.9%)
Avg confidence on corrected image	74.1%
Corrected-to-0 rate (orig correct)	100.0%
Corrected-to-0 rate (orig wrong)	72.7%
Error detector signal	STRONG

When the original prediction was correct, 100% of corrected images are predicted as 0° . When wrong, only 72.7% are — the 27.3% gap is a reliable signal to flag potential errors without ground truth labels.

7. Production Recommendation

Based on these findings, the following inference strategy is recommended:

Step	Action	Trigger
1	Run standard inference	Always
2	Check confidence $\geq 75\%$	Always
3	If $<75\%$: rotate to all 4 orientations, re-infer, pick highest confidence	Only $\sim 6\%$ of images
4	Rotate-to-0 consistency check as final validation	Optional — adds 1 extra inference call

Cost: Retry triggers on $\sim 6\%$ of images. Each retry costs 4 $\times$ inference. Net overhead: $\sim 24\%$ more inference calls overall ($0.06 \times 4 = 0.24$). At 34ms per call (ONNX), avg latency per image = $34 \times 1.24 = \mathbf{42ms}$ — well within document preprocessing SLA.

Gain: Rescues 7 of 11 uncertain errors. The remaining 4 uncertain errors and 6 high-confidence errors require fine-tuning to fix, not retry logic.

Rescue rotation cheat sheet (production use):

If model predicts...	Best rescue rotation	Avg gain
0° (uncertain)	+270° or +180°	+8–9pp
90° (uncertain)	+90°	+19.7pp (100% cases)
180° (uncertain)	+180°	+28.6pp (88% cases)
270° (uncertain)	+90° or +270°	+18–20pp